

Temporal consistency of EEG signals in affective video annotation

Master: Intelligent Systems

Subject: SJK016 – Master Thesis (2023/2024)

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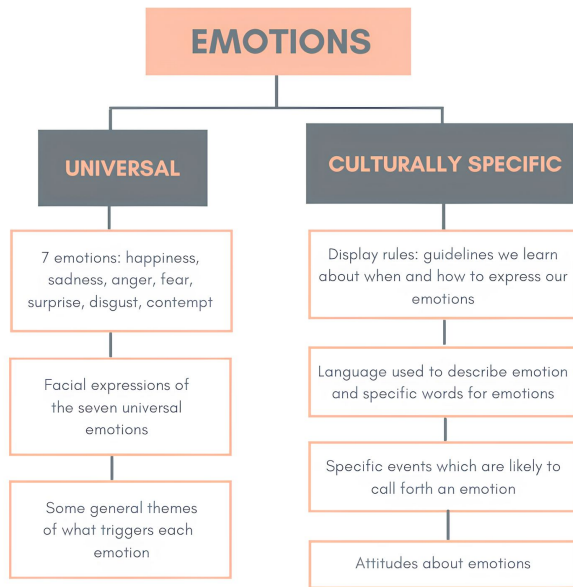
Conclusions

01

Introduction

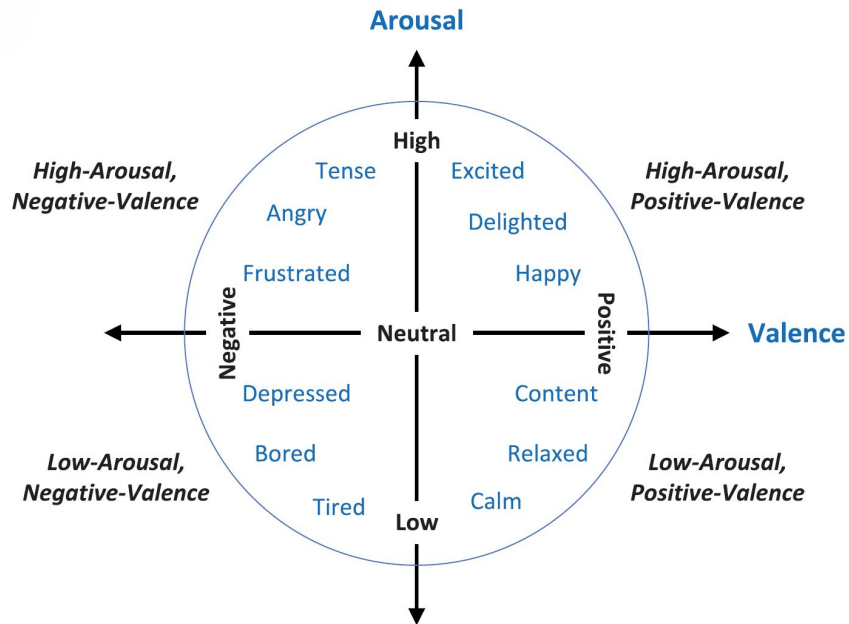


Emotions in Affective Computation



Paul Ekman Group

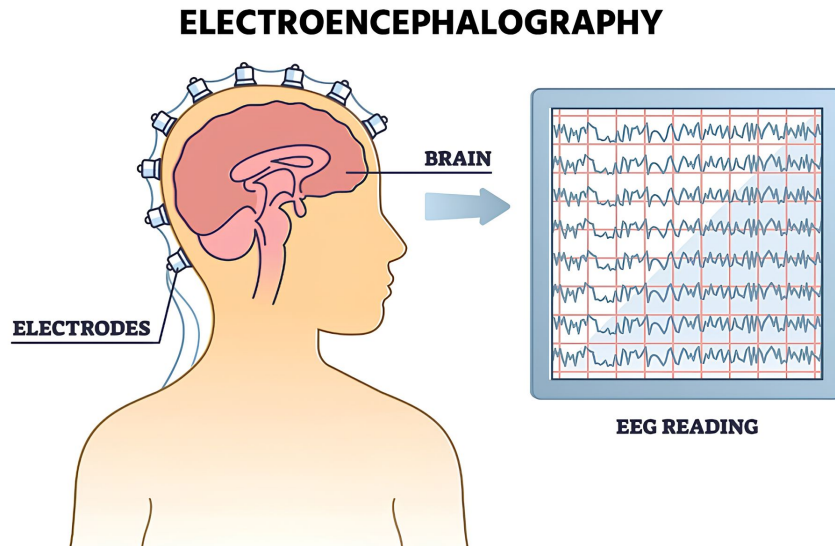
(Ekman, 1992)



Circumplex Model (Russell, 1980)

Electroencephalography (EEG)

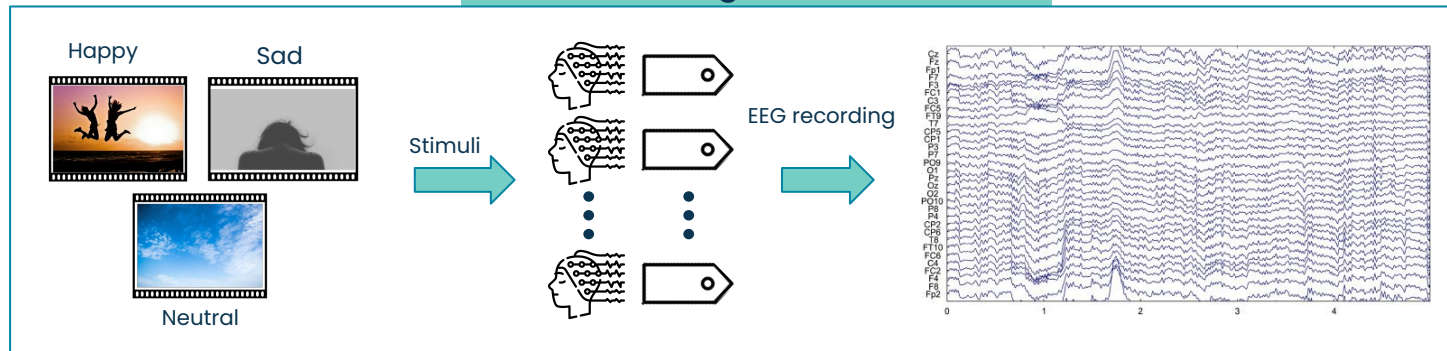
"EEG is an electrophysiological technique for the recording of electrical activity arising from the human brain" (NIH, 2016).



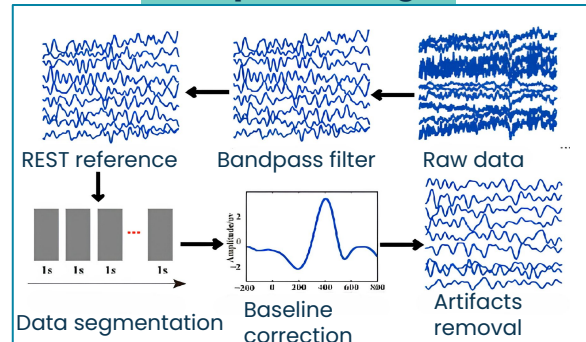
Complex interplay between physiological signals and emotional experiences.
(James-Lange, 1885) and (Cannon-Bard, 1915)

EEG Emotion Recognition

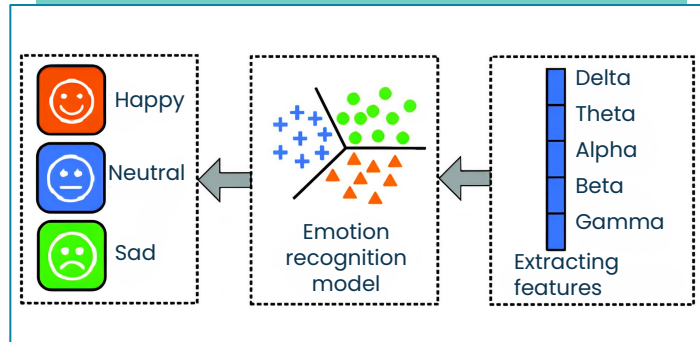
Crowdsourcing + Data Collection



Preprocessing



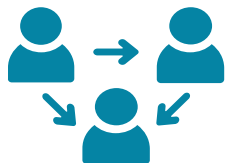
Feature Extraction and Classification



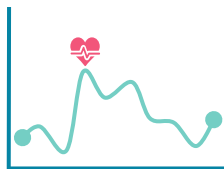
Open Problems



Feature Selection and Extraction



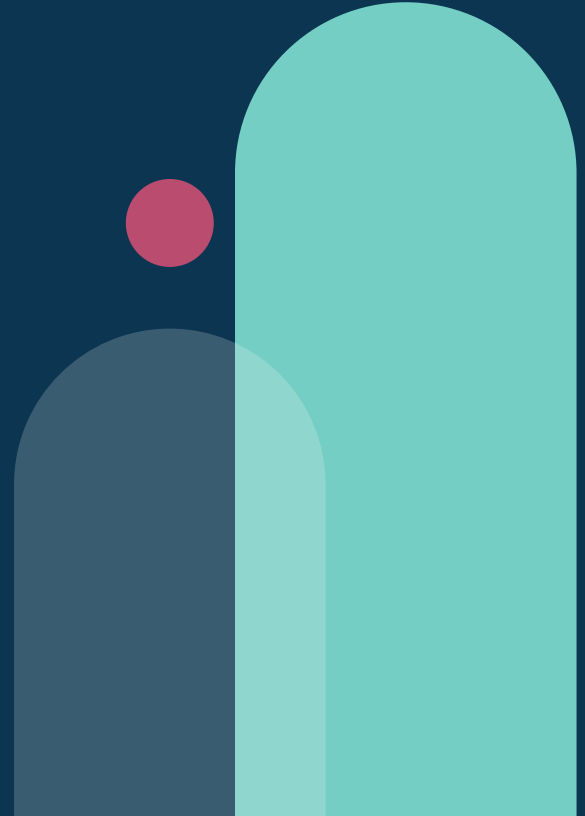
Inter-Subject Variability



Affective Consistency

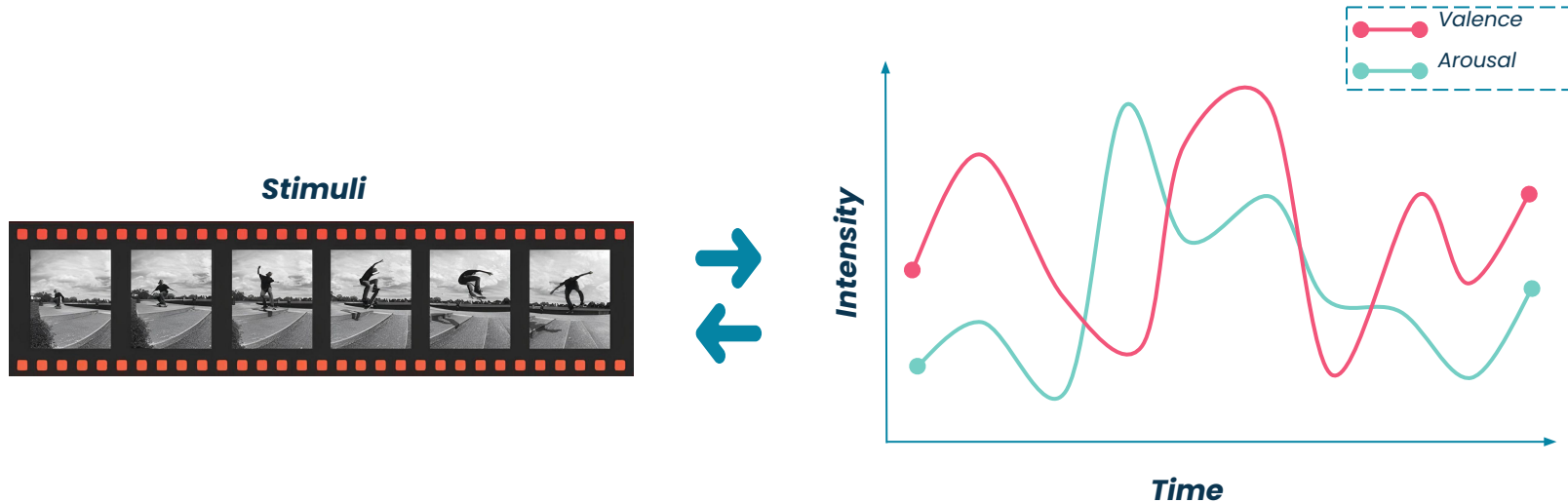
02

The Problem

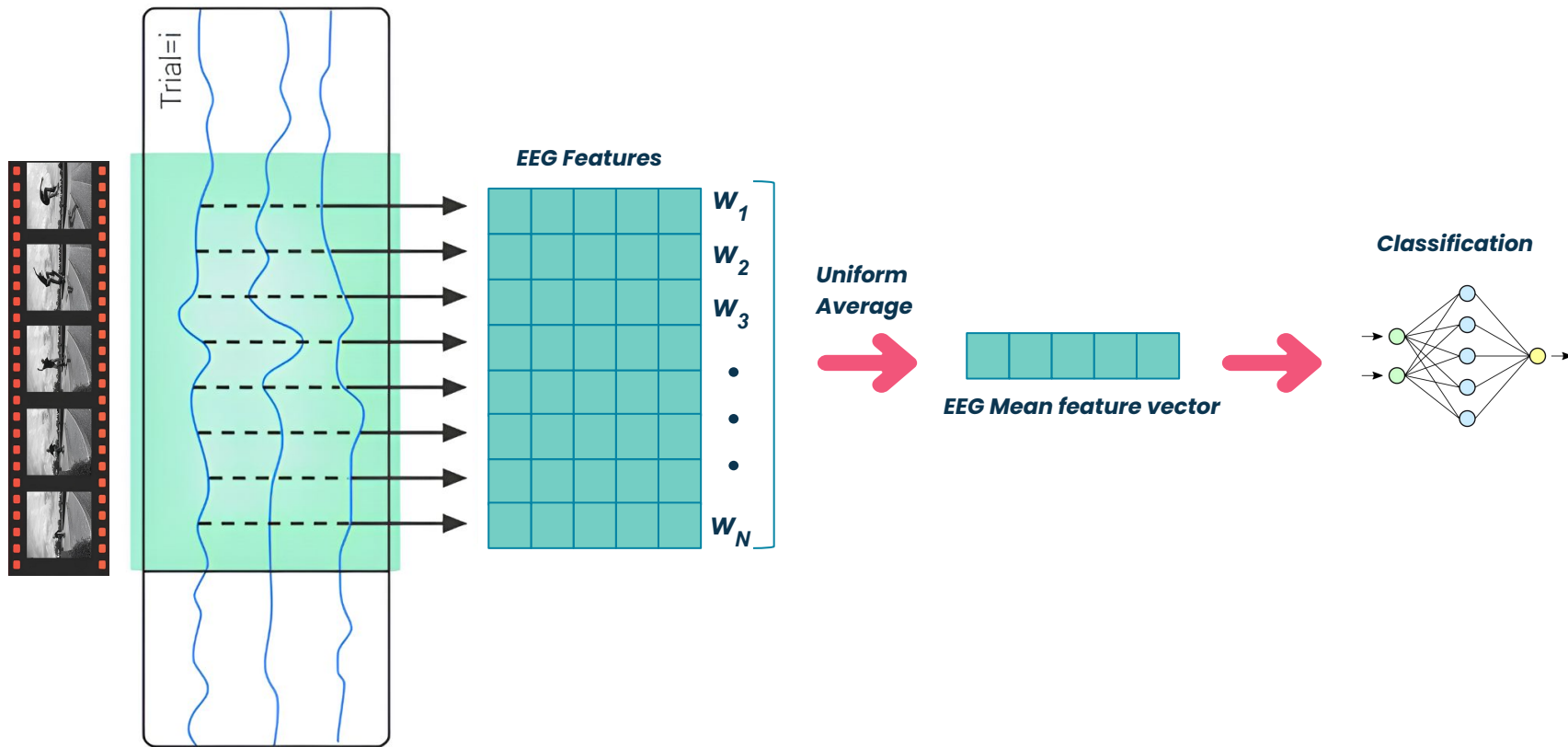


Affective Consistency

Coherent association between emotional states and their corresponding physiological response patterns within subjects.

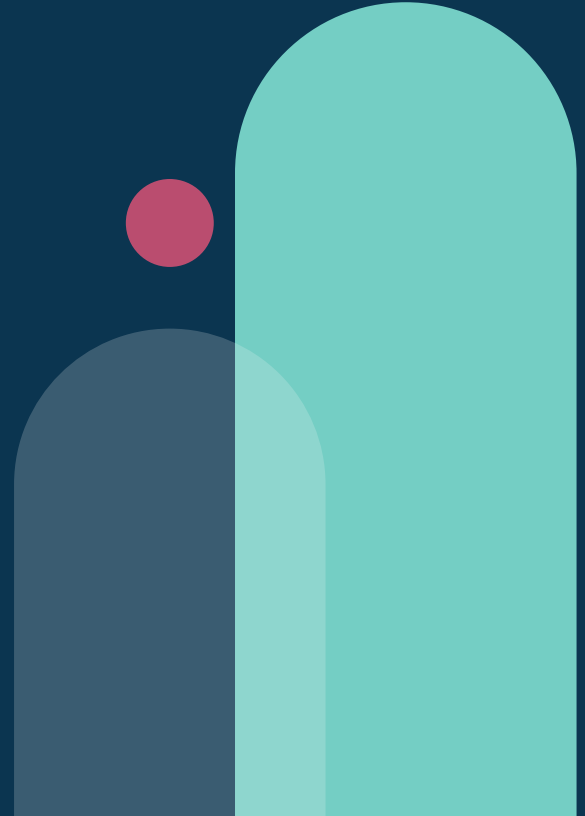


Temporal Affective Consistency



03

Research Questions



Research Questions



RQ1

Temporal Affective Consistency Scores



RQ2

Consistency-weighted › Uniform



RQ3

Improve Generalizability



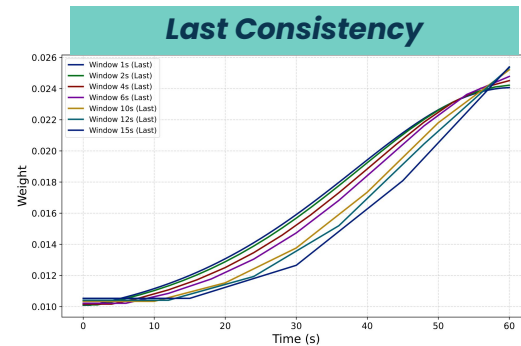
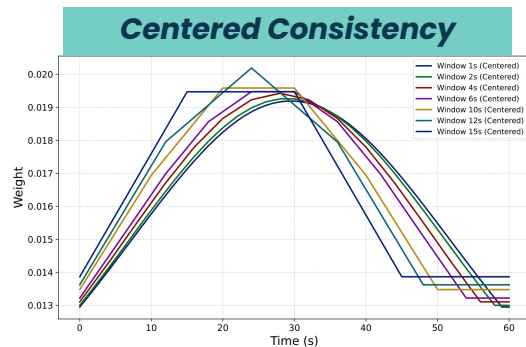
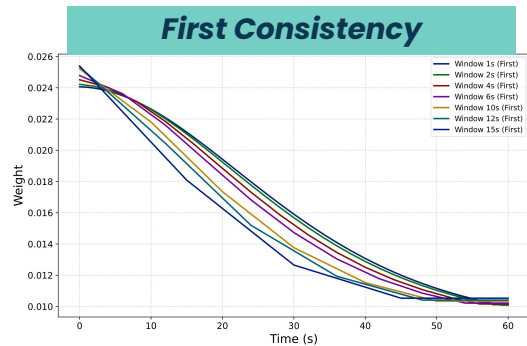
04

Methodology

- Consistency Scores
- Datasets
- Algorithms
- Evaluation Protocols
- Qualitative Analysis

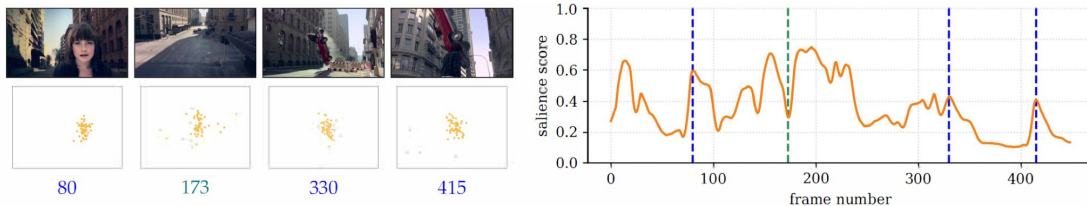


Heuristic Consistency



Data-Aware Consistency (i)

$$\textit{Glimpse} \quad s(t) = \frac{2}{n(n-1)} \sum_{\substack{i,j \in \{1, \dots, n\} \\ i \neq j}} \mathbb{1}[d_{ij} < \theta_s], \quad t \in \{1, \dots, T\},$$



Adaptation to EEG

$$\textit{EEG-Glimpse} \quad a^v(t) = \frac{2}{n(n-1)} \sum_{\substack{i,j \in \{1, \dots, n\} \\ i \neq j}} \mathbb{1}[d_{ij}^v < \theta^v], \quad t \in \{1, \dots, T_\omega\},$$

- d_{ij}^v Distance of feature vectors of subjects i and j for the video v
- θ^v Video-dependent threshold

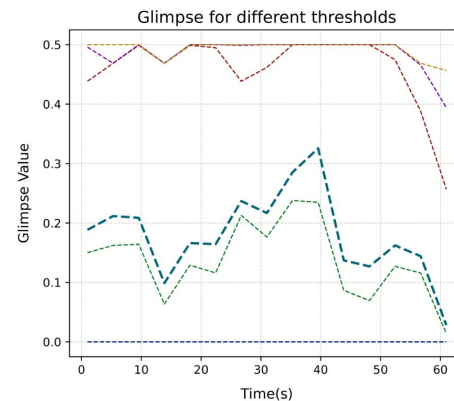
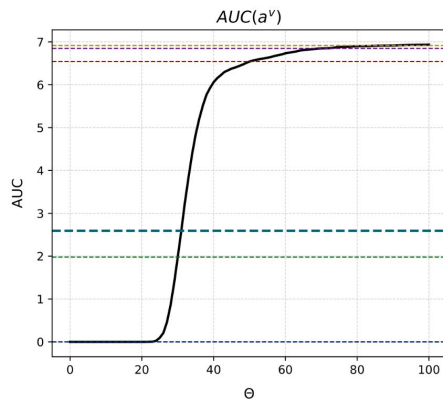
Data-Aware Consistency (ii)

Threshold Selection

Maximize capture of common emotional responses



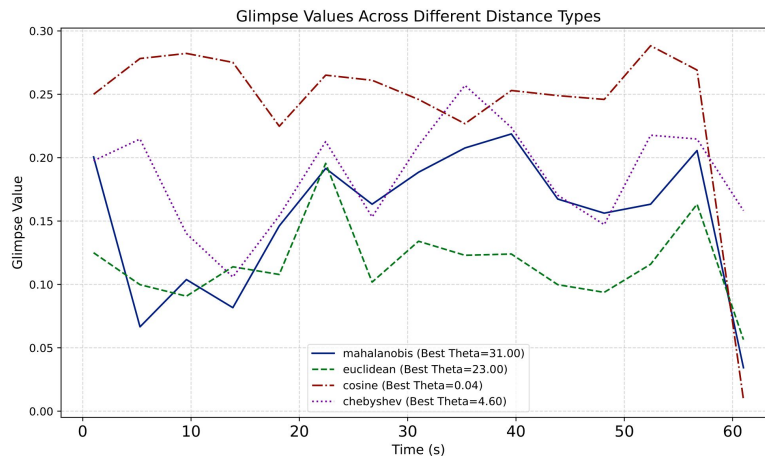
Minimize capture of different emotional responses as common



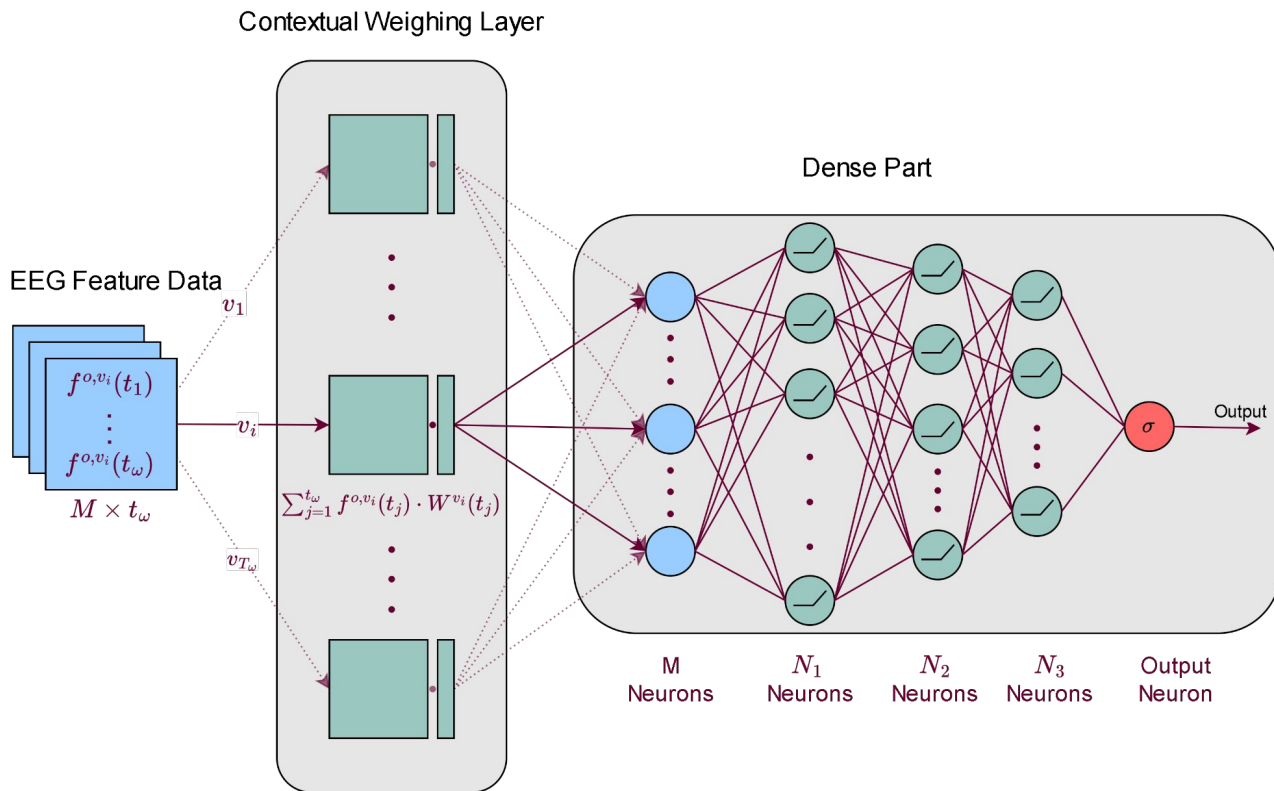
Distance Selection

Mahalanobis Distance

$$d(x, y) = \sqrt{(x - y)^T S^{-1} (x - y)}$$



Learning-Based Consistency



Datasets



DEAP Dataset

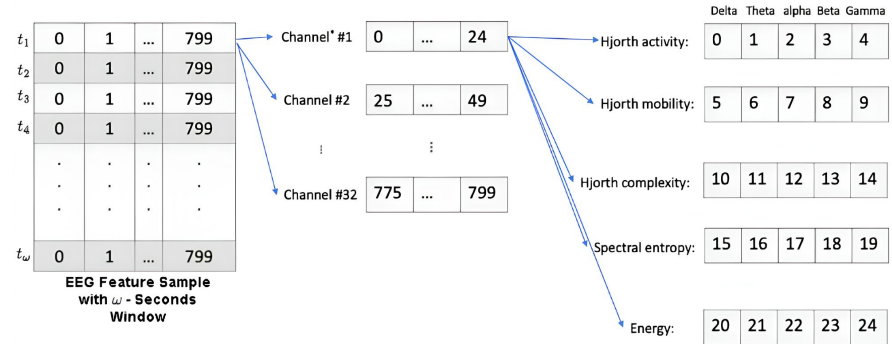
- 32 Participants
- 40 videos of 60s length
- Binary Valence and Arousal Classification
- 512 Hz signal through 32 channels



MAHNOB-HCI Dataset

- 27 Participants
- 20 videos between 35s and 117s
- Binary Valence and Arousal Classification
- 1024 Hz signal through 32 channels

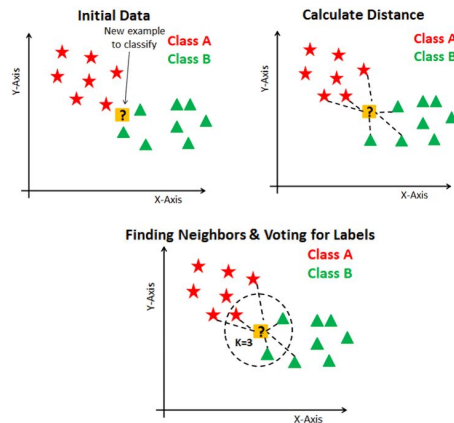
EEG Feature Sample



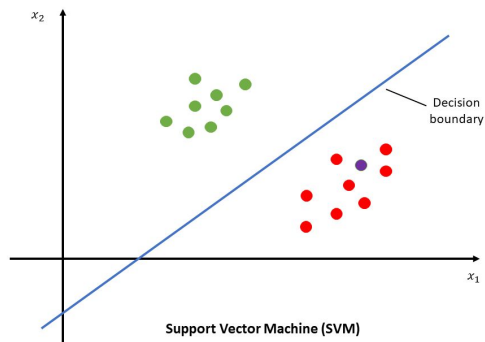
Provided by the Banana research group

Algorithms

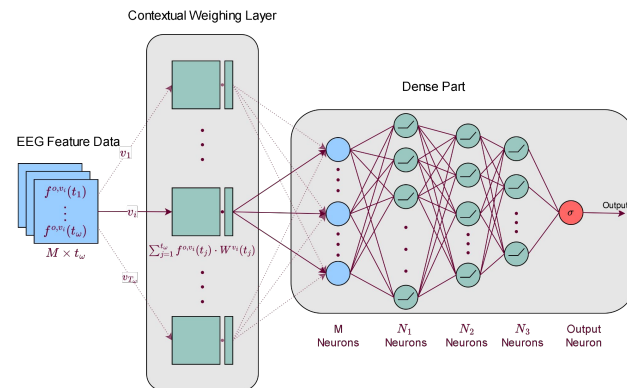
K-Nearest Neighbours



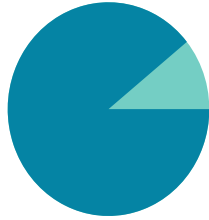
Support Vector Machine



EEG Contextual Neural Network



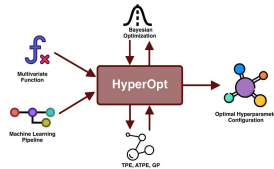
K-Fold Cross Validation



90-10 Stratified Sampling

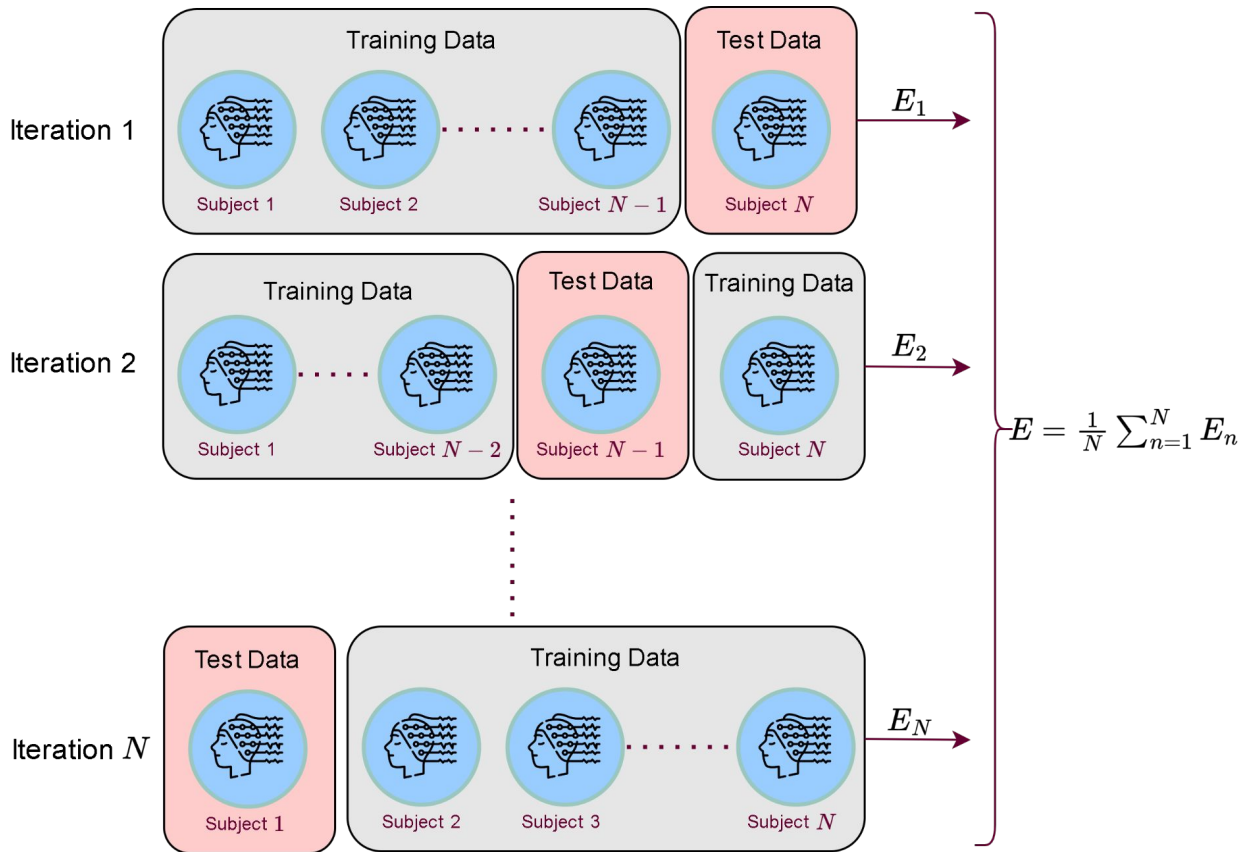


10-Fold Cross Validation

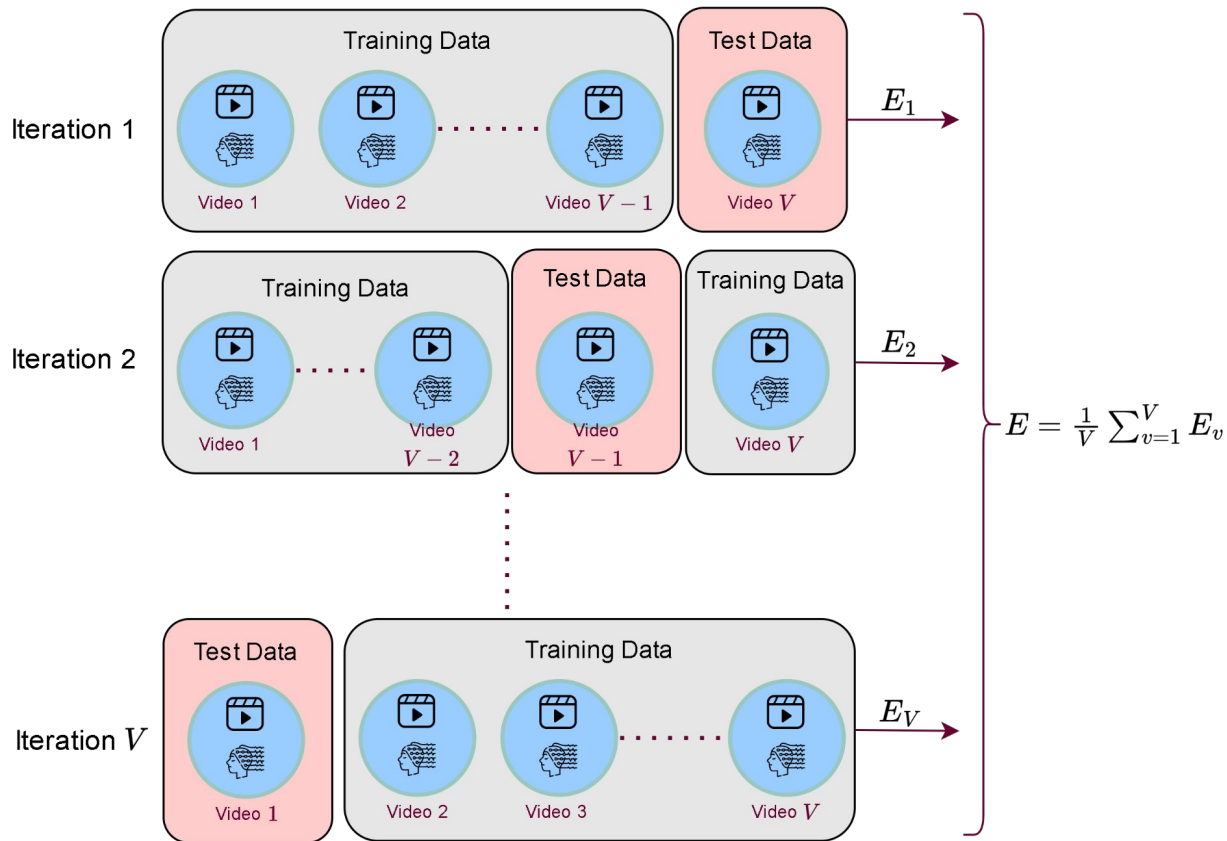


Hyperparameter Optimization

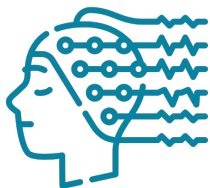
Leave-One-Subject-Out (Loso)



Leave-One-Video-Out (Lovo)



Subject-Dependent Model (SDM)



EEG Data of One Subject



LOVO within Subject's Data



Repeated over subjects

05

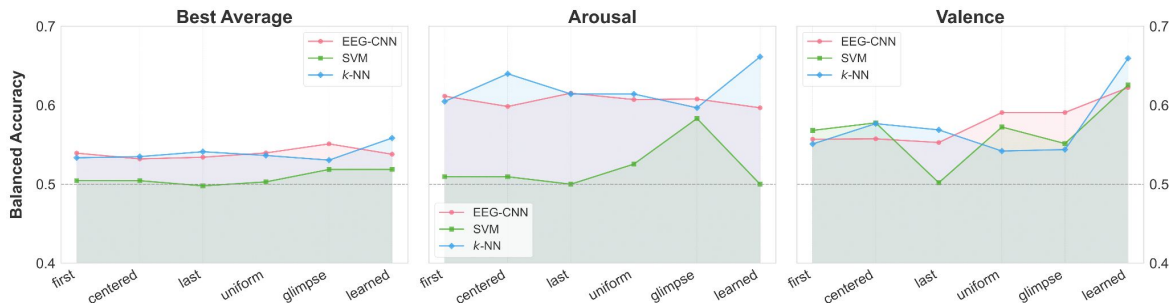
Consistency Results



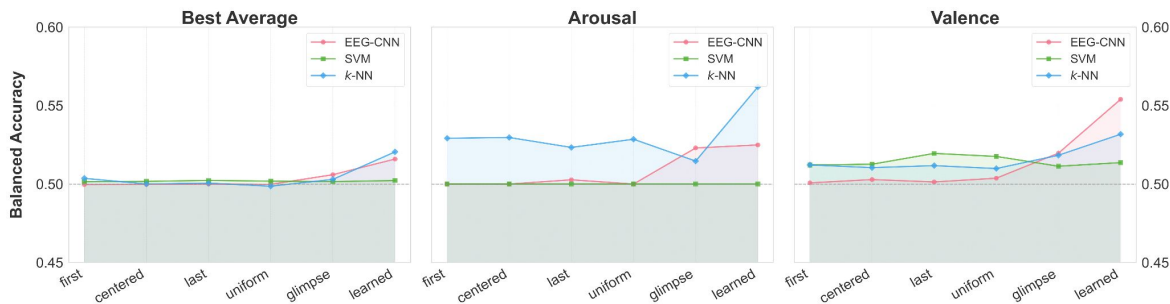
Consistency Impact on Emotion Recognition

DEAP

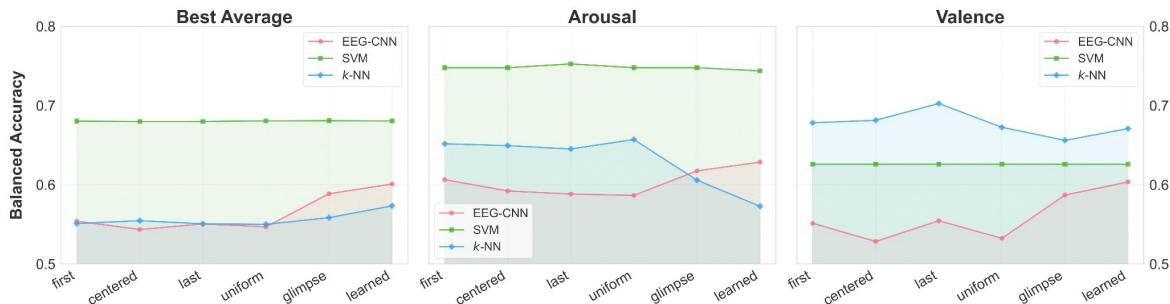
K-Fold



LOSO



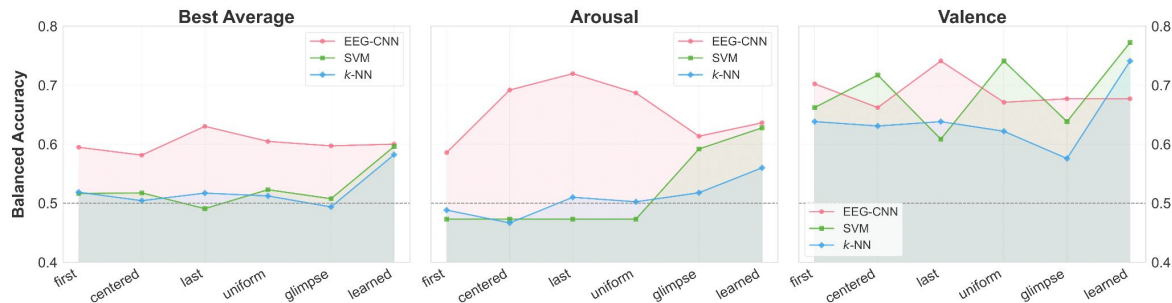
SDM



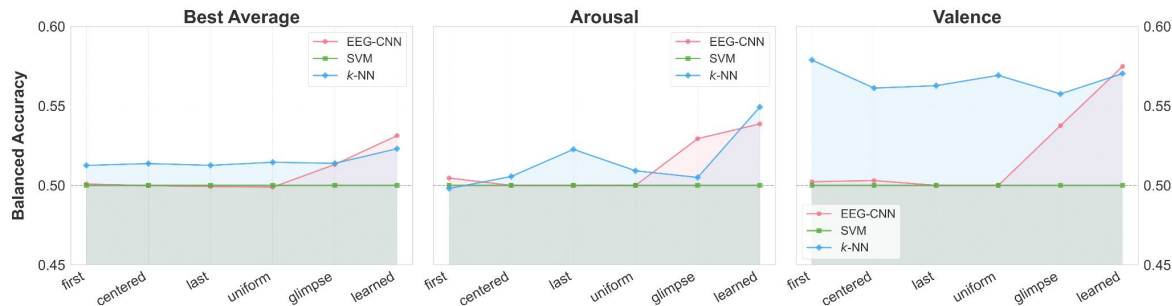
Consistency Impact on Emotion Recognition

MAHNOB-HCI

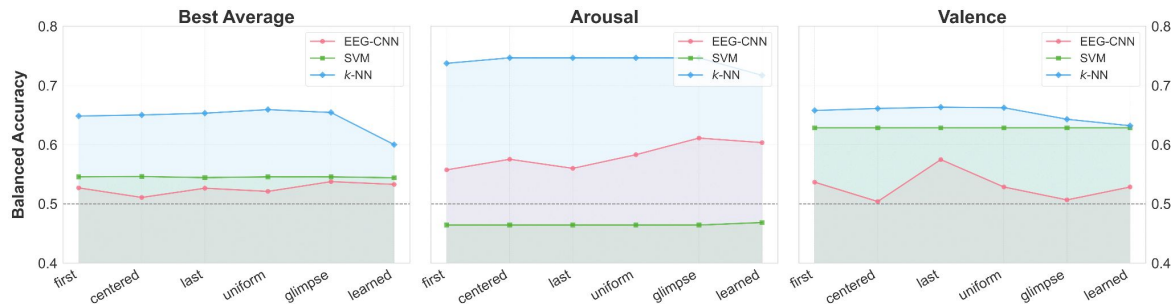
K-Fold



LOSO



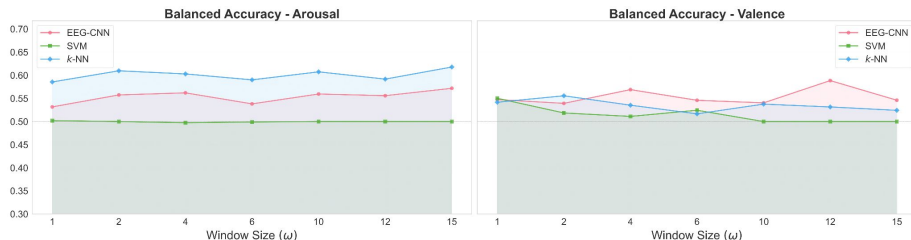
SDM



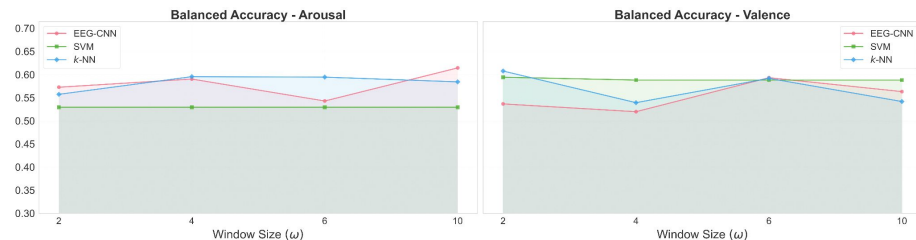
Consistency Impact on Emotion Recognition

LOVO Results

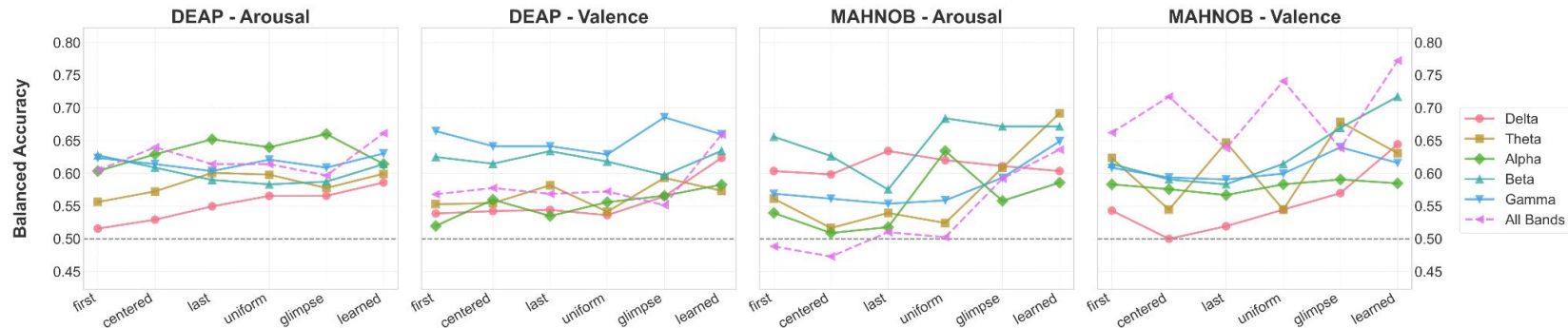
DEAP



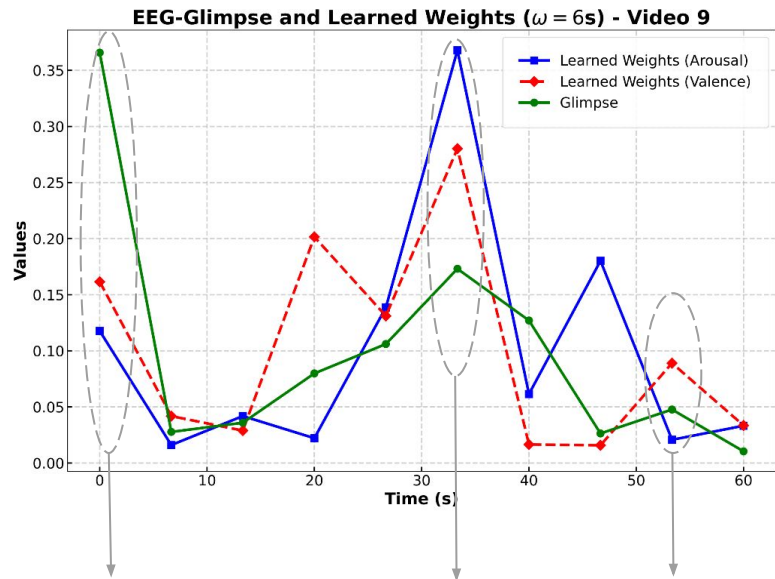
MAHNOB



Per-Band Analysis



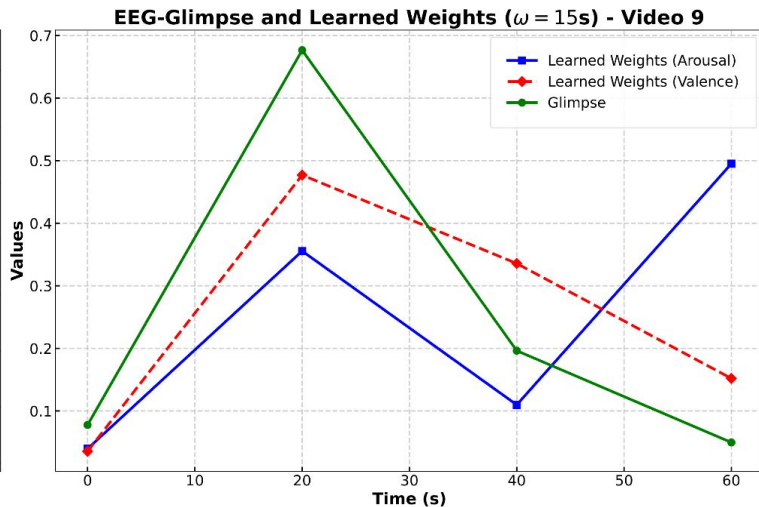
Consistency Agreement Analysis (i)



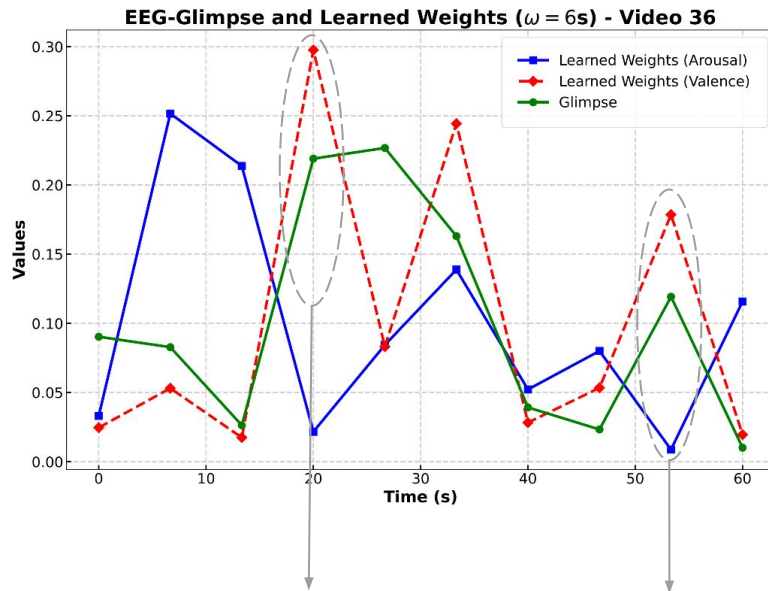
Familiar Song
Starts

Music Starts

Vocal Melody Starts

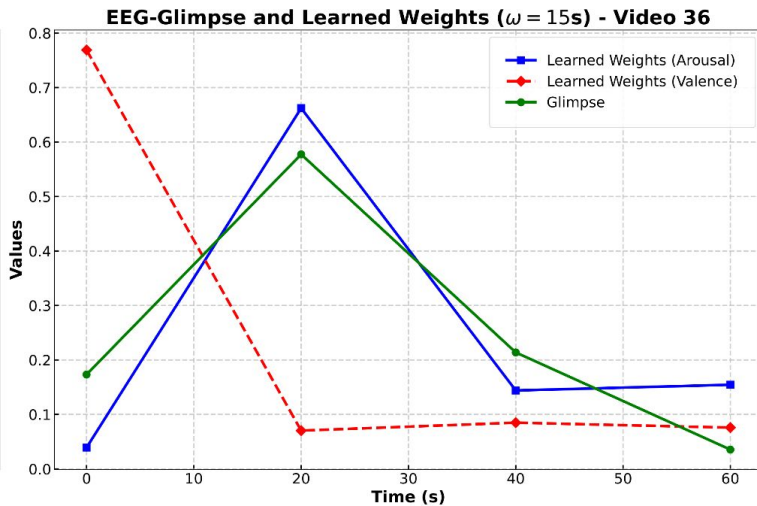


Consistency Agreement Analysis (ii)



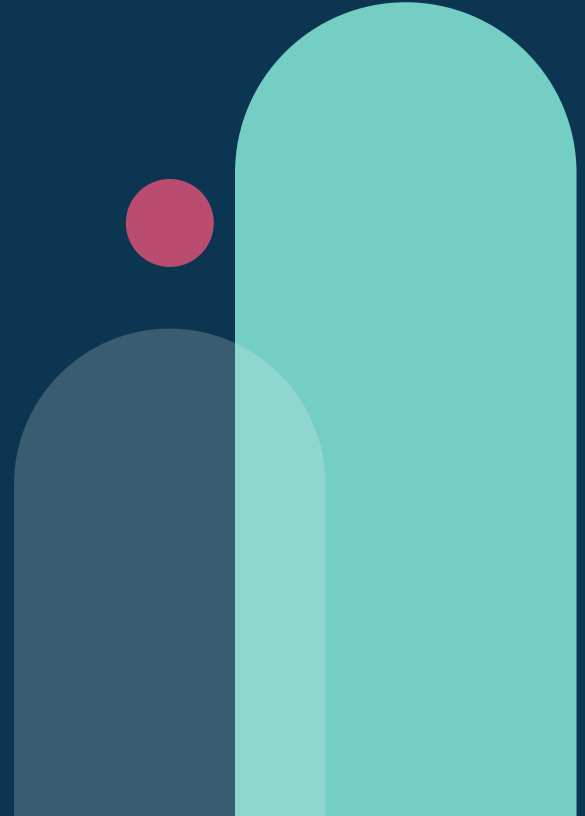
Impactful Images

Music Starts



06

Conclusions



Summary Findings

F1

Data-aware and Learning-based scores could reliably quantify temporal affective consistency

F2

Consistency-weighted >> Uniform

F3

Consistency-Weighted Improve Generalizability

Thanks!

Do you have any questions?

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